

## ***Machine Learning Open Ended Project Description***

**Total score: 100**

**Note:** This project is to be completed collaboratively as a team. You **may use AI tools** (e.g., ChatGPT, GitHub Copilot) to assist with understanding concepts, generating helper code, developing app interfaces, or checking grammar and spelling. However, **all submitted work must be written by your team** and reflect your collective understanding. If you use AI tools during the project, you **must clearly document how they were used** in your final report (e.g., “Used ChatGPT to help debug the gradient descent implementation” or “Used Copilot to generate initial layout code for the Streamlit app”).

In this term project, you are asked to identify a real-life engineering problem that can be solved by machine learning and associated dataset. You then need to perform machine learning tasks, such as classification and regression etc., to the datasets, and report your results and observations.

### **Research Procedures:**

#### **Identify engineering problem to solve and suitable datasets**

In order to identify a suitable dataset to use, start with an engineering domain that interest you. Examples include predictive maintenance, fault detection in mechanical systems, energy consumption forecasting, traffic flow analysis, sensor-based activity recognition, and process optimization in manufacturing. There are many public available datasets available. We list a collection of datasets as follows but you are encouraged to use search engines to identify your own datasets that interest you. Be prepared to spend substantial data in pre-processing and exploring the datasets that you choose.

Kaggle ( HYPERLINK "https://www.kaggle.com/datasets" <https://www.kaggle.com/datasets>)

Amazon wishlist data ( HYPERLINK "http://www.applefritter.com/bannedbooks" <http://www.applefritter.com/bannedbooks>)

MIT Lincon DARPA intrusion detection evaluation data ( HYPERLINK "http://www.ll.mit.edu/IST/ideval/data/data\_index.html" [http://www.ll.mit.edu/IST/ideval/data/data\\_index.html](http://www.ll.mit.edu/IST/ideval/data/data_index.html))

ImageNet ( HYPERLINK "http://www.image-net.org/" <http://www.image-net.org/>)

Various dataset in KDD cup ( HYPERLINK "http://www.kdnuggets.com/datasets/kddcup.html" <http://www.kdnuggets.com/datasets/kddcup.html>)

UCI KDD achieve ( HYPERLINK "http://kdd.ics.uci.edu/" <http://kdd.ics.uci.edu/>)

Wikipedia dump ([http://en.wikipedia.org/wiki/Wikipedia:Database\\_download](http://en.wikipedia.org/wiki/Wikipedia:Database_download))

KDnuggets datasets ( HYPERLINK "http://www.kdnuggets.com/datasets/index.html" <http://www.kdnuggets.com/datasets/index.html>)

U.S. Geological Survey FIPS55 data ( HYPERLINK "http://geonames.usgs.gov/domestic/download\_data.htm" [http://geonames.usgs.gov/domestic/download\\_data.htm](http://geonames.usgs.gov/domestic/download_data.htm))

**What to turn in? The final report for this section should contain the following components:**

An introduction of the engineering problem that you are interested in

A description of the dataset that you selected. It should include details such as the origin and size of the datasets, how the data is represented, e.g. graph, records, attribute, statistics of the values of the attributes...

Describe any exploratory analysis and results that you have performed on your datasets. The raw and processed datasets.

A formal problem definition with what is given, what is the goal, and what are the constraints.

A plan for your machine learning tasks

**Related Work**

In this phase you will need to identify and summarize the work that has been done and is similar to yours. Try resources such as google scholar ( HYPERLINK "http://scholar.google.com" <http://scholar.google.com>), citeseer ( HYPERLINK "http://citeseer.ist.psu.edu/" <http://citeseer.ist.psu.edu/>), DBLP ( HYPERLINK "http://www.informatik.uni-trier.de/~ley/db/" <http://www.informatik.uni-trier.de/~ley/db/>), ACM Digital Library ( HYPERLINK "http://portal.acm.org/dl.cfm" <http://portal.acm.org/dl.cfm>) or IEEE Explorer ( HYPERLINK "http://ieeexplore.ieee.org/Xplore/guesthome.jsp" <http://ieeexplore.ieee.org/Xplore/guesthome.jsp>) to *identify at least two papers that use the dataset or techniques related to yours*. Write a summary of the findings. Make sure to put your work into context.

**What to turn in? The related work section of your final report should contain the following components:**

A summarization of the literature

A discussion on the difference of your approach from the techniques described in the papers you reviewed if any.

**Perform machine learning tasks on your dataset**

In this phase, you will try the intended machine learning tasks on your dataset. Use a language you are confident with (Matlab or Python is recommended) to write your scripts/codes (*you can directly use built-in functions from standard libraries*).

Each team member is required to implement **at least one machine learning algorithm that was not covered in class**. For example, if your team has two members, the team must implement at least two different algorithms, each contributed by a different member. Clearly specify which member implemented which algorithm in the “Individual Contributions” section of your report.

In addition, you must **compare these new algorithms to at least one algorithm taught in class**. Analyze the results, evaluate performance, and discuss the strengths and weaknesses of each method.

### **What to turn in? A report describing**

The methods implemented and any other tasks that you performed.

Your experimental results.

Comparison of the experiment results of different method(s).

Possible explanations for the experiment results.

Discussions of the strengths and weaknesses of each method.

### **Develop Application and User Interaction**

In this phase, you are required to build a **simple interactive application or interface** that demonstrates **one** of the machine learning models you implemented. The purpose of this component is to help you explore how machine learning models can be used in practical, user-facing settings.

You may use any tool or framework you are comfortable with, such as:

Python-based frameworks (e.g., Streamlit, Flask, Gradio, Jupyter Notebook with widgets)

MATLAB tools (e.g., App Designer, Web App Server)

Basic web development tools (e.g., HTML/CSS/JavaScript)

Your app should:

Allow the user to input or upload data

Display the result or prediction generated by your machine learning model

Provide basic explanation or visualization of the output if possible (e.g., confidence score, classification label, or plotted result)

You only need to demonstrate one of your machine learning models in the app. You may choose the model that performs best or the one you believe is most suitable for deployment.

You are encouraged to use AI tools (e.g., ChatGPT, GitHub Copilot) to assist with the app development process. For example, you may use them to generate boilerplate code, help debug, or suggest user interface designs. However, the app must reflect your own understanding, and you must clearly explain how AI tools were used in your report.

### **Conclusion**

Summarize what you did and what you have learned from this machine learning task.

Describe any future work you think is worthy while.

### **Video Demonstration (10 points)**

You are required to record a short video (1-3min) demonstrating your project. Your video should:

Include a brief overview of the project goal

Demonstrate how the application works (e.g., input, output, interaction)

Highlight key results and conclusions

You may record the video using any screen recording tool (e.g., Zoom, Loom, OBS, PowerPoint, etc.). Please upload your video to a platform such as YouTube, Google Drive, or OneDrive, and include the link in your final report.

### **Final Report (85 points)**

The expected length of your final report is 5-7 pages (single column, 1 inch margins, 11 font size).

The final report should include all contents identified in Section I and follow the outline below:

#### **Introduction (5 points)**

(An introduction of the application domain)

#### **Related Work (7 points)**

(In this phase you will need to identify and summarize the work that has been done and is similar to yours. Should include:

A summarization of the literature (5 points)

A discussion on the difference of your approach from techniques described in the papers you reviewed (2 points)

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#### **Problem Formulation (10 points)**

(A formal problem definition with what is given, what is the goal, and what are the constraints.)

#### **Methods (10 points)**

(Description of ML methods implemented.)

### **Data Description and Experimental Setup (10 points)**

(First, describe the dataset and include all details such as the origin and size of the datasets, how the data is represented, e.g. graph, records, attribute, statistics of the values of the attributes...

Then, describe the experimental environment and the settings of your experiments, e.g., how you choose the training dataset, the values of the model parameters, etc.)

### **Experimental Results and Analyses (20 points)**

(Describe the experimental results and your observations and insights gained from the results.)

### **Application and User Interaction (10 points)**

(Describe the design of your application and its features.)

### **Conclusion (3 points)**

(Summarize the problem solved in this project, the methods used, and the results obtained.)

### **Video Demo (10 points)**

(Provide the link to your video)

### **Reference**

**Individual contributions (must be included, otherwise the report will not be graded)**

(Describe the contribution of each member in the team. If you do not have teammates, you do not need to include this section.)

### **Proposal Presentation (5 points)**

Prepare slides (3-5) for a 3-minute presentation about your project idea and your plan to implement this idea. **You will need to present your proposal during the class.**

### **Final Presentation (10 points)**

Prepare slides (10-15) for a 10-minute presentation about what you did. The slides should summarize the content of your final report. **You will need to present your project during the class.**

### **Final Submission**

**A zip file that includes the following items:**

All codes

Final report

Proposal presentation slides

Final presentation slides

One team only needs to submit once.